ISOKO — Audit y’urubuga

Raporo y’ukuri: ibibazo, ibigomba gukosorwa uyu munsi, no kugereranya n’izindi mbuga

Itariki: 14 Ugushyingo 2026 · Repo: eddyizere24-collab/Isoko- (branch main)

1. Igisubizo kigufi

Urubuga rufite umusingi mwiza kurusha uko benshi batangira, ariko si rwiteguye gufungurirwa rubanda uyu munsi. Ibyo byubatswe neza ni byinshi (reba igice cya 3). Icyakora hari ibibazo bike bikomeye bishobora gutuma umucuruzi wa mbere ananirwa gutangira, cyangwa abantu babi (abazamura imibare y’ibinyoma, abuzuza database) bagahungabanya urubuga mu minsi mike.

Icya mbere kigaragara kuri screenshot yawe: ubutumwa «Ongera winjire muri konti yawe» ku ntambwe ya 1 yo kongeraho igicuruzwa. Ni ikibazo nyacyo mu kode (reba #1). Umucuruzi yuzuza ifishi, yagera ku ifoto agahagarara. Ibyo byonyine byasenya ikizere cy’abacuruzi ba mbere.

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2. Aho nashingiye (n’ibyo ntashoboye gukora)

Nasomye: kode yose y’ingenzi muri repo (server.ts, src/lib/api.ts, neon.ts, pages z’umucuruzi, MarketplaceContext), imigrati ya database (SQL) 004, 005, 007, 010, 013, 016, Neon Function y’amafoto, CI, render.production.yaml, n’inyandiko zanyu zo muri docs/. Nasomye kandi screenshot yawe.

Ntashoboye:

- Gufungura urubuga live: umuyoboro w’urubuga wanze ko robot ifungura urupapuro (robots). Rero **ibyo mvuga ni ibyo nabonye muri kode**, si ibyo nagerageje ku rubuga ruri hanze.
- Gukora signup/upload nyayo, cyangwa gusoma database ya Production. Ibyo biracyari «ibizamini bikenewe» (nk’uko inyandiko yanyu ubwayo ibivuga).
- Kubona amakuru yizewe kuri «Mazu»: ntabonye isoko ryizewe. Kugereranya ni na Jumia, Amazon, n’imbuga z’u Rwanda nabonye (reba igice cya 6).

Aho ntemeza 100% nanditse «bigomba kugenzurwa». Ntabeshye.

3. Ibyo mwakoze neza — mubirinde

- **Umutekano wa database wubatse neza:** RLS ikoze ku mameza yose; abacuruzi ntibashobora guhindura plan, product_limit cyangwa status y'iduka (migration 007 yabyambuye uburenganzira); kongeraho/guhindura/gusiba ibicuruzwa bica muri RPC zigenzura nyirabyo.
- **Ukuri ku kimenyetso cya «Verified»:** ikimenyetso kigaragara gusa iyo identity_status = verified. Ntimubeshya abaguzi. Ibi ni bike ku mbuga nyinshi.
- **Nta data y'ibinyoma:** demo accounts, KYC z'ibinyoma n'amafoto y'icyitegererezo byakuweho (PR #7). Iduka ritangira ari ubusa kandi rivugisha ukuri.
- **Kwiyandikisha byoroshye:** email, WhatsApp n'izina ry'iduka gusa. Nta ndangamuntu itangwa mbere. Ibi bihuye n'abacuruzi bato.
- **Amafoto:** gusuzuma ubwoko bw'ifoto mu mutwe w'idosiye (magic bytes), kugabanya ingano (500 KB), gukuraho GPS, gukoresha ububiko bw'ibanga ku bya KYC.
- **Igare rigabanywa ku mucuruzi** n'ubutumwa bwa WhatsApp bwateguwe; kugenzura igiciro n'ububiko igihe igare rifunguwe.
- **CI ifite ubushishozi:** TypeScript, unit tests, smoke tests kuri Neon Setup na Production, build, byose kuri buri PR.
- **Ikinyarwanda** n'amagambo y'ishakisha (nayiki → nike, ijini → ipantalo ya jean) ni ikintu Jumia/Amazon badafite — ni imbaraga yanyu.
- **Kubungabunga ukuri mu nyandiko:** «Security Gate» ivuga neza ibitarageragezwa. Ibyo ni ubunyangamugayo buke ku mishinga nk'iyi.

4. Ibibazo byose byabonetse

Uburemere: **Gikomeye cyane** = guhagarika gufungura; **Gikomeye** = mbere yo gutangaza; **Hagati** = icyumweru cya 1; **Gito** = isuku.

A. Gikomeye cyane (gukosora uyu munsi)

#	Uburemere	Ikibazo (ahantu)	Icyo gukora
1	**Gikomeye cyane**	**Kwinjira kw'ikinyoma (ikibazo cya screenshot yawe).** Ubutumwa «Ongera winjire muri konti yawe» buturuka muri getNeonJwt() (src/lib/neon.ts) iyo Neon nta token ifite. Ariko urupapuro rwo kongeraho (SellerAddProductPage) rwizera gusa authSession ibitse muri localStorage ikimenyetso «managed-neon-auth». Muri MarketplaceContext, iyo icyo kimenyetso kihari, nta kongera kugenzura session nyayo (restore irahagarara) kandi ntigisibwa keretse Neon ivuze «signed_out». Rero umucuruzi abona ko yinjiye, agakora intambwe nyinshi, agahagarara ku ifoto. Impamvu zishoboka (nta kimenyetso gihagije): session yarangiye; cyangwa mushakisha ya telefoni ihagarika cookies zo kuri domaine ya gatatu, kuko Auth iri kuri neonauth…neon.tech naho urubuga ruri kuri onrender.com.	Mbere yo kwerekana ifishi, genzura token nyayo (neon.auth.token()); niba ibuze, siba ikimenyetso, wohereze kuri /injira. Kora «ikizamini cy'ubushakashatsi» kuri telefone 3 zitandukanye (Chrome, Samsung Internet, Opera). Niba ari cookies, shyira Auth na API kuri subdomaine imwe n'urubuga.
2	**Gikomeye cyane**	**Nta kizamini cyuzuye (end-to-end) cyigeze gikorwa.** «Security Gate» ivuga ko: signup, session, upload y'ifoto, guhindura ububiko n'ubugenzuzi bitarageza n'umucuruzi nyawe. Ikibazo #1 ni ibimenyetso by'iryo shyiramo ry'ibitarageragejwe.	Kora ikizamini kimwe cyuzuye ku rubuga ruri hanze: kwiyandikisha → ongeraho igicuruzwa n'amafoto 3 → kigaragare → ugure kuri WhatsApp. Ongeraho Playwright muri CI.

#	Uburemere	Ikibazo (ahantu)	Icyo gukora
3	**Gikomeye cyane**	**Nta backup yigenga.** Inyandiko yanyu ubwayo: backup yo hanze ya Neon ntiyashyizweho, kandi restore ntirageregeza. Gusubiza inyuma kwa Neon Free ni kugufi (reba igenamiterere ryanyu). Niba ikosa rya migration cyangwa konti ihagaritswe, amaduka n'ibicuruzwa byose bishobora kubura.	Shyiraho backup ya buri munsi (pg_dump) ibitse ahandi + kopi y'amafoto. Gerageza restore rimwe mbere yo gufungura.

B. Gikomeye (mbere yo gutangaza kuri rubanda)

#	Uburemere	Ikibazo (ahantu)	Icyo gukora
4	**Gikomeye**	**Ubwishyu bw'ububiko bw'amafoto bufite igisenge.** Amafoto atangwa na Neon Object Storage ubwayo. Inyandiko yanyu ivuga: **5 GB y'urujya n'uruza ku kwezi**. Ifoto ya 300 KB × 17 000 kureba = 5 GB. Urubuga rukunzwe rwarenga icyo gisenge mu minsi mike, amafoto agahagarara.	Shyira Cloudflare (ubuntu) imbere y'amafoto, cyangwa ugure igice cyishyurwa. Shyiraho alert ku gukoresha.
5	**Gikomeye**	**Urubuga rukurura icyegeranyo cyose cy'ibicuruzwa icyarimwe.** fetchBootstrap() isoma v_public_bootstrap (amaduka + ibicuruzwa byose) buri gihe urubuga rufunguwe; ishakisha ribera muri mushakisha. Iduka rimwe rishobora kugira ibicuruzwa 500. Kuri telefone ya make n'internet y'umuvuduko muto, iyo JSON izaba nini cyane. (Sinasomye ibyo view ikata — bigomba kugenzurwa.)	Shyiraho paginasiyo ku ruhande rwa server (amapaji 24), ishakisha rikorwe muri database (Postgres full-text), amafoto thumbnail gusa kuri lisiti.
6	**Gikomeye**	**Imibare ishobora kubeshywa (views, clicks, cart_adds).** track_marketplace_event_v1 ikoresha session_id itangwa na mushakisha ubwayo; umuntu ahindura session_id akongera views uko ashaka. track_campaign_v1 nta dedupe na gato ifite. Kandi **popularity_7d** ikoreshwa mu ntonde z'ishakisha — umucuruzi yazamura ibicuruzwa bye ku buryo bw'ikinyoma.	Bara ku ruhande rwa server (IP + user-agent hash), ushyireho rate limit, kandi uhagarike gukoresha popularity mu ranking kugeza ikingiwe.
7	**Gikomeye**	**Database ishobora kuzuzwa n'abanyabyaha.** Ameza event_dedup, report_rate, subscribers yandikwa n'abatazwi (anonymous) nta rate limit nyayo, kandi nta cleanup. Neon Free ifite 1 GB gusa; iyo yuzuye, abacuruzi nabo ntibashobora kwandika.	Ongeraho cron isiba event_dedup ya kera (iminsi 2), rate limit ya IP, na Turnstile/CAPTCHA kuri signup, raporo na subscribe.
8	**Gikomeye**	**CAPTCHA ni ikinyoma.** submitReport yakira turnstileOk ariko ntiyigeze iyohererereza RPC. Rate limit ya raporo ishingiye kuri session_id y'ubusa, ihinduka byoroshye. Umuntu yuzuza ikarita y'ubuyobozi raporo nyinshi.	Shyiraho Cloudflare Turnstile nyayo ugenzurwa na server, cyangwa ukureho ubutumwa buvuga CAPTCHA.
9	**Gikomeye**	**Amategeko ari kuri client gusa.** Itegeko «nta nimero ya telefoni/link mu izina cyangwa ibisobanuro», n'izina rya 5–60, bigenzurwa muri validation.ts gusa. RPC create_seller_product_v1 yemera izina rya 2–120 kandi ntisuzuma telefoni/link. Umuntu ukoresha API ataciye mu ifishi arenga amategeko. Ndetse path_full y'ifoto ishobora kuba URL yo hanze cyangwa ifoto y'undi mucuruzi (publicAssetUrl yemera http(s) iyo ariyo).	Subiramo amategeko yose muri SQL (CHECK / RPC). Emera gusa amafoto afite prefix «products//».

#	Uburemere	Ikibazo (ahantu)	Icyo gukora
10	Gikomeye	**Buto ya Google irerekanwa kandi ntigeze ishyirwaho.** LoginPage yerekana «Komeza ukoresheje Google», ariko «Security Gate» ivuga ko OAuth ya Google nta client ID/secret ifite kandi ngo «nta buto yerekanwe». Kandi signInWithGoogle ijyana kuri /abacuruzi?google=1; ikibazo cya Google account itarafite profile gikemurwa na «provisional session» (reba #12).	Cyangwa shyiraho credentials za Google ugerageze, cyangwa ukureho buto kugeza zihari. Nongere nguhe ukuri: ntabwo nashoboye kuyigerageza.
11	Gikomeye	**Konti ya Admin ntishobora gusubizwa.** Admin yiyandikisha na nimero@login.isoko.invalid. Aho si email nyayo, rero «Wibagiwe ijambo ry'ibanga» ntikora kuri we, kandi nta 2FA. Admin umwe ni we rukuta rw'urubuga rwose; niba yibagiwe ijambo ry'ibanga, abura byose. Kandi /admin/setup irahari kuri rubanda; umutekano wayo ushingiye ku kode imwe (nta rate limit ya kode nasanze).	Bifatire Admin email nyayo, ushyireho 2FA, kandi uhindure kode ya mbere nyuma yo kuyikoresha (irasibwa, byiza). Ongeraho rate limit kuri claim_first_admin_v1.

C. Hagati (icyumweru cya mbere)

#	Uburemere	Ikibazo (ahantu)	Icyo gukora
12	Hagati	**«Provisional session».** Iyo Neon ifite user ariko nta profile (urugero Google), MarketplaceContext yandika localStorage uwo muntu nk'umucuruzi. Kandi allowAnonymous: true muri neon.ts bishobora gutuma na bamwe batinjiye bagira user.id. Ibi bishobora kwerekana menu y'umucuruzi ku muntu utari we (ntibimuha uburenganzira, byitiranya gusa).	Garagaza «provisional» nk'ubusa; ohereza kuri /abacuruzi kurangiza kwiyandikisha. Genzura niba anonymous user atera iki kibazo.
13	Hagati	**Nimero ya telefoni itemejwe ishobora «gufatwa».** Itegeko «PHONE_ALREADY_REGISTERED» rituma umuntu wiyandikishije mbere na nimero y'undi (nta OTP) amubuza kwiyandikisha. Kandi nta kintu kibuza umuntu gukoresha WhatsApp y'undi mucuruzi nk'iye.	Ongeraho OTP ya WhatsApp/SMS, cyangwa ubutumwa bwo kwemeza nimero (MTN/Airtel).
14	Hagati	**Imeri yo kugarura ijambo ry'ibanga ntiyageragejwe.** Neon Auth ikoresha SMTP ihuriweho (auth@mail.myneon.app), kandi email ntiyemezwa mu kwiyandikisha. Imeri irashobora kujya muri spam cyangwa gutinda, kandi umuntu ashobora kwiyandikisha n'email atari iye.	Shyiraho SMTP yawe (domaine yawe), uhagarike konti zitaremeza email ku bikorwa bikomeye.
15	Hagati	**Amafoto asigara mu bubiko iyo ibicuruzwa bisibwe.** Buri foto yoherezwa kabiri (full + thumb) mbere yuko igicuruzwa kibikwa. Iyo umucuruzi ahagaritse, cyangwa agasiba igicuruzwa, amafoto nta wuyasiba (function ifite upload gusa). Kandi umucuruzi umwe afite igisenge cy'ibicuruzwa 500 × amafoto 5 × 2 × ≤500 KB = **kugeza kuri 2.5 GB mu bubiko bw'ubuntu bwa 5 GB** (ikiruta kose).	Ongeraho gusiba amafoto, cron isiba imfubyi, kandi ugabanye igisenge ku bicuruzwa 50–100 mu kwezi kwa mbere.
16	Hagati	**Nta rate limit/quota ku mafoto.** Function ya upload ntiyigera ibuza umucuruzi wemewe kohereza amafoto adashira.	Igisenge ku munsi kuri buri konti.

#	Uburemere	Ikibazo (ahantu)	Icyo gukora
17	Hagati	**Shops: social_links na amagambo atagenzurwa kuri server.** Ukwemera https gusa ni muri updateSellerShop (client). Umuntu ukoresha Data API ataciye mu ifishi yashyiraho URL mbi; shyira CHECK muri SQL. Kandi about/landmark nta gisenge cya DB.	Genzura muri SQL (CHECK cyangwa trigger) hanyuma wemeze uko ShopPage yerekana links (rel=noopener).
18	Hagati	**Neon Free: gusinzira kwa database.** Compute iguma isinziriye nyuma y'iminota 5; ubusabe bwa mbere burarenza kwihuta. Urubuga rwerekana hero vuba (neza), ariko ibicuruzwa bitegereza.	Shyiraho skeleton ihamye, «nongera ugerageze» ku mashusho, kandi utekereze igice cyishyurwa cya Neon.
19	Hagati	**Amategeko y'u Rwanda (Law 058/2021).** Amakuru nk'email na telefoni ni ay'umuntu ku giti cye. Inyandiko nabonye zivuga: kwiyandikisha muri NCSA nk'ushinzwe amakuru (data controller), ububiko mu Rwanda keretse hari icyemezo, no kumenyesha NCSA mu masaha 48 iyo amakuru yibwe. Database yanyu iri Frankfurt (aws-eu-central-1).	Baza umunyamategeko cyangwa NCSA. Ibi si inama z'amategeko, ni ukwibutsa. Politiki y'ibanga (/ibanga) igomba gusobanura ayo makuru.

D. Gito / isuku (iyo hari umwanya)

#	Uburemere	Ikibazo (ahantu)	Icyo gukora
20	Gito	**Kode ya kera ikiri muri repo:** server.ts (47 KB) ni Express y'icyitegererezo cya AI Studio. Ntikoreshwa muri Production (urubuga rukora nka static site), ariko npm start iracyayiyobora. Ifite session token ikomoka kuri uid:id:hash(id+passwordHash) itarangira kandi ikorwa na hashPassword; ntabwo yigeze iba umutekano. Iyo umuntu abyitiranya akayishyira ku murongo, byaba isenyuka.	Siba server.ts, src/server/, src/data (iyo itakoreshwa), supabase/ (imigrati ya kera y'undi mushinga), .env.example (GEMINI_API_KEY), n'amapaki adakoreshwa (@google/genai, express).
21	Gito	**Inyandiko zivuguruzanya:** ISOKO_HOSTING_DECISION.md ivuga Cloudflare Pages kandi ivuga Render «idakwiye»; Production iri kuri Render static. CHANGELOG ivuga «local JSON db» na «nta git», ibyo byakuweho. Bulk add: docs bivuga 10, commit ivuga 50.	Vugurura docs muri PR imwe: «Aho turi ubu».
22	Gito	**Ihuriro ry'imigrati rirabuze.** Imigrati iri ahantu habiri (supabase/ na infra/neon/), ishyirwaho ukuboko, nta rollback. Inyandiko ya 013 yashyizemo URL ya storage ya Production mu view (016 yaravuguruye).	Koresha umuyoboro umwe wa migrations muri CI, kandi usuzume drift hagati ya repo na Production.
23	Gito	**Ipaki ya beta ku bintu bikomeye:** @neondatabase/neon-js 0.9.0-beta ni yo ikoreshwa kuri Auth. Beta ishobora guhinduka nta kuburira. Package name ni «react-example».	Bika versiyo, usome changelog mbere ya buri update, kandi uhindure izina.
24	Gito	**Repo ni public:** SQL, amategeko yose y'umutekano, n'amazina y'ibikoresho (project IDs) bigaragara. RLS irinda, ariko abanyabyaha basoma uburyo bwose.	Tekereza kuyigira private (GitHub Free ibyemera).
25	Gito	**Nta domaine yawe.** onrender.com ntiyizerwa ku bacuruzi n'abaguzi, cyane cyane ku mafaranga; kandi cookies z'imbuga zitandukanye (#1) ni ikibazo.	Gura domaine (urugero isoko.rw / isoko.co.rw) hanyuma ushyireho Auth/API kuri subdomaine yawe.

5. Gereranya n’izindi mbuga

Aho amakuru ari ku isoko ryizewe nashyizeho isoko. Ibya Amazon ni ubumenyi rusange; ntabwo nayigenzuye uyu munsi. **Kuri «Mazu» ntabonye isoko ryizewe**, bityo sinavuga ibyayo.

Ibintu	ISOKO ubu	Jumia	Amazon	Ibyo mwakwigiraho
Kwemeza umucuruzi	Email + WhatsApp. Nta kwemeza. Iduka rihita ryemerwa.	KYC: indangamuntu/pasiporo, TIN, bank account; igenzura rifata iminsi 2–3 y’akazi (Vendor Hub)	Kwemeza umuntu n’ikarita/banki mbere yo gucuruza	Gukomeza byoroshye ni byiza, ariko iduka rishya ryerekanwe «ritaremezwa» kandi ugire urwego rwa «Ryemejwe» rufite inyungu.
Kwishyura	Nta na kimwe: umuguzi n’umucuruzi bumvikana kuri WhatsApp.	Jumia ifata amafaranga, ikishyura abacuruzi buri cyumweru	Amazon ifata ubwishyu kandi irinda umuguzi	MoMo/Airtel Money ni ngombwa mu gihe kizaza; ubu mwirinde kwemeza ko urubuga «rurinda» ubwishyu.
Kurinda umuguzi	Raporo + guhisha ibicuruzwa. Nta isubizwa/ububiko bw’ibirego.	Politiki yo gusubiza, serivisi y’abakiriya	Garanti y’umuguzi, gusubiza, ubuhamya	Tangira n’amategeko yanditse neza yo gusubiza no gukemura ibirego; byorohereza ikizere.
Ubuhamya / ratings	Nta na kimwe.	Hari	Hari kandi ni urufatiro rw’ikizere	Shyiramo ubuhamya bw’abaguzi nyuma y’amezi make; ubu kwerekana «imyaka umucuruzi amaze» ni inzira y’agateganyo.
Gutanga (delivery)	Inyandiko y’umucuruzi (≤120 inyuguti).	Ifite logistics/ububiko	Ifite FBA/ububiko bwayo	Ntimukwiye kubikora byose; shyiramo uduce tw’aho gutanga (akarere/umurenge) kuboneka mu ishakisha.
Ishakisha	Ikinyarwanda + synonyms; ribera muri mushakisha.	Server-side	Server-side, ranking ikomeye	Ishakisha ryanyu ry’Ikinyarwanda ni imbaraga. Rishyirwe kuri server mbere yuko ibicuruzwa biba byinshi.
Imbuga za nyuma ntoya mu Rwanda	Iduka rya WhatsApp.	Iduka Online ryishyuza ifatabuguzi + ikiguzi cya delivery; ryavuze ko kwemeza abacuruzi gucuruza kwari ingorabahizi; RwandaMart ifashwa na GIZ/MINICOM; Lulu.rw ishyira imbere kugenzura kubera uburiganya; Yubeyi yavuze ko kutizera kwishyura ukoresheje ikarita ari ikibazo gikuru.		Ikibazo gikomeye si ikoranabuhanga; ni ugutuma abacuruzi n’abaguzi bizera. Tangirana n’abacuruzi 10–20 nyabo mu karere kamwe.

Ukuri kuzuye: ISOKO ubu ni «catalogue y’amaduka ihuza abaguzi na WhatsApp», si «marketplace» nka Jumia/Amazon. Ibyo ntibibi, kandi biroroshye guhera aho. Icyaba kibi ni ukwita urubuga ibyo rutari byo. Mwirinde amagambo nk’«ubwishyu bwizewe» cyangwa «amaduka yose yemejwe».

6. Gahunda y’ibikorwa

Uyu munsi (mbere y’umucuruzi wa mbere)

- **#1** Gukosora kwinjira kw’ikinyoma: kugenzura token nyayo mbere y’ifishi; kohereza kuri /injira; gusiba ikimenyetso cya kera.
- **#2** Gukora ikizamini kimwe cyuzuye kuri telefone 3 (kwiyandikisha → igicuruzwa n’amafoto → kigaragare → WhatsApp).
- **#10** Gukuraho buto ya Google kugeza ishyizweho kandi igeragejwe.
- **#3** Gutangira backup ya buri munsi ibitse hanze ya Neon (n’ikizamini kimwe cya restore).
- **#11** Guha Admin email nyayo, 2FA, no gukingira /admin/setup.

Mbere yo gutangaza kuri rubanda

- **#6, #7, #8** Rate limit na Turnstile nyayo; gusiba event_dedup ya kera; guhagarika popularity mu ranking.
- **#9** Amategeko yo kuri client agomba no kuba muri SQL; amafoto yemerwa gusa niba ari aya nyirayo.
- **#4, #5, #15** Cloudflare imbere y’amafoto, paginasiyo, no gusiba amafoto y’imfubyi; igisenge cy’ibicuruzwa 50–100 mu kwezi kwa mbere.
- **#13, #14** SMTP yawe, no kwemeza nimero.
- **#25** Domaine yawe.
- Politiki y’ibanga n’amabwiriza byanditse mu Kinyarwanda neza; kugenzura Law 058/2021 (#19).

Nyuma yo gufungura

- Gukora isuku (#20–#24): gusiba server.ts na supabase/, kuvugurura docs.
- Ubuhamya bw’abaguzi, imibare y’umucuruzi, na MoMo igihe cyose mubishoboye.
- Gutangira n’abacuruzi 10–20 bazwi mu karere kamwe, mubabwira ibyo mushaka kumenya mbere yo gufungura rubanda.

7. Ibyo ntabashije kugenzura

- Urubuga live (robots yanze); uko rukora kuri telefone zitandukanye; kuboneka kwa Google OAuth; kuba Admin yarashyizweho.
- Ibikubiye muri v_public_bootstrap (ingano ya JSON) n’umuvuduko nyawo.
- Ibikubiye muri HomePage, SellerApplyPage, ShopPage, SellerProductsPage, AdminPage, n’imigrati 006, 008, 009, 011, 012, 014, 015 — ntabwo nazisomye zose.
- Imiterere nyayo ya Neon Auth (cookies, anonymous, SMTP) kuko ntabwo nashoboye kugera ku madashibodi ya Neon cyangwa Render.

8. Inkomoko

- Kode yanyu: github.com/eddyizere24-collab/Isoko- (commit f13db971), docs/ISOKO_PRODUCTION_SECURITY_GATE.md, docs/ISOKO_HOSTING_DECISION.md.
- Jumia Vendor Hub: vendorhub.jumia.sn (Verification iminsi 2–3 y’akazi, KYC); vendorhub.jumia.ug/kyc-documents-mandatory.
- Disrupt Africa: «Rwandan startup launches e-commerce marketplace platform» (Iduka Online); «Rwanda’s Yubeyi tapping in…» (kutizera ikarita).

- GIZ: «Rwanda: Enabling the Future of E-commerce» (RwandaMart); AllAfrica: «Rwanda: Buy, Sell and Connect With Lulu» (Lulu.rw).
- Law 058/2021: DLA Piper Data Protection (Rwanda); PKF Rwanda Data Protection Alert (Oct 2023); PwC Rwanda «Why data protection in Rwanda goes beyond registration» (Kamena 2026).